

Silybum marianum syn. *Carduus marianus* (Asteraceae)

Milk Thistle, Mary thistle

Milk thistle has been used in Europe as a remedy for depression and liver problems for hundreds, if not thousands, of years. Recent research has confirmed traditional herbal knowledge, proving that the herb has a remarkable ability to protect the liver from damage resulting from alcoholic and other types of poisoning. Today, milk thistle is widely used in the West for the treatment of a range of liver conditions.



Milk thistle has distinctive white markings on its leaves caused, according to tradition, by the Virgin Mary's milk.

Habitat & Cultivation

Native to the Mediterranean, milk thistle grows throughout Europe and is naturalized in California. It grows wild, thriving in open areas, and is cultivated as an ornamental plant. Milk thistle likes a sunny position and self-seeds readily. The flower heads are picked in full bloom in early summer and the seeds are collected in late summer.

Related Species

Other closely related herbs, including holy thistle (*Cnicus benedictus*, p. 191) and globe artichoke (*Cynara scolymus*, p. 199), protect the liver from toxicity and exert a positive restorative action on liver function.

Key Constituents

- Flavonolignans (1–4%)
(known collectively as “silymarin”)
- Bitter principles
- Polyacetylenes

Key Actions

- Liver protective
- Chemoprotective

- Anticancer
- Anti-allergenic
- Increases breast-milk production

Research

■ **Silymarin** German research from the 1970s onward has focused on silymarin, a substance contained in the seeds. This exerts a highly protective effect on the liver, maintaining its function and preventing damage from compounds that are normally highly toxic. It has been shown that severe liver breakdown, resulting from ingesting carbon tetrachloride or death cap mushrooms, can be prevented if silymarin is taken immediately before, or within 48 hours. In Germany, silymarin has been used successfully to treat hepatitis and liver cirrhosis.

■ **Other research** In several clinical trials, silymarin extracts protected liver function in people taking chemotherapy for cancer; including, in one trial, children suffering from leukemia. Research so far suggests that silymarin extracts support healthy liver function during chemotherapy and do not reduce the effectiveness of chemotherapy drugs. Laboratory research indicates that silymarin, particularly silybin, has significant anticancer properties. In a 2011 clinical trial in Iran, patients with allergic rhinitis experienced significantly fewer symptoms when taking a silymarin extract.

Traditional & Current Uses

■ **Traditional uses** Milk thistle flower heads, boiled and eaten like artichokes, were useful as a spring



Milk thistle

A spiny biennial, growing to 5 ft (1.5 m), with white-veined leaves and purple flower heads.

Parts Used

Flower heads are eaten as a tonic food and can be used in remedies.



Fresh flower head

Dried flower head

Spiny, thistle-like leaves are gray-green



Seeds

Seeds contain silymarin, which protects the liver. They are the main part used in remedies.

Key Preparations & Their Uses

⚠ **Cautions** May cause allergic reactions in rare cases. Seek guidance from your health care practitioner if planning to take alongside chemotherapy.



Decoction of seeds (to make, p. 291). For liver infections, take ½ cup (75 ml) a day.



Tincture of seeds may be taken to help hay fever.

Capsules of seeds (to make, p. 292). For a hangover, take a 500 mg capsule.



🌿 **Tablets** are prescribed for long-term treatment of liver disorders.

tonic after the winter months when people had been deprived of fresh vegetables. They were also taken to increase breast-milk production, and were considered excellent for *melancholia* (depression), which was traditionally associated with the liver. Gerard states in his *Herball* of 1597, “My opinion is that this [milk thistle] is the best remedy that grows against all melancholy diseases.”

■ **Liver disorders** Today, milk thistle is the main remedy used in Western herbal medicine to protect the liver and its many metabolic activities, and help renew its cells. The herb is used in the treatment of hepatitis and jaundice, as well as in conditions where the liver is under stress—whether from infection or excess alcohol, or from chemotherapy.